

Problem1

May 3, 2026

0.1 2.1 Cart-pole swing-up with limited actuation

(a)

The convex approximation (LOCP) around the current iterate $(s^{(i)}, u^{(i)})$ is

$$\begin{aligned} \text{minimize}_{s, u} \quad & (s_N - s_{\text{goal}})^\top P (s_N - s_{\text{goal}}) \\ & + \sum_{k=0}^{N-1} [(s_k - s_{\text{goal}})^\top Q (s_k - s_{\text{goal}}) + u_k^\top R u_k] \\ \text{subject to} \quad & s_0 = \bar{s}_0 \\ & u_k \in \mathcal{U}, \quad \forall k \in \{0, \dots, N-1\} \\ & \|u_k - u_k^{(i)}\|_\infty \leq \rho, \quad \forall k \in \{0, \dots, N-1\} \\ & \|s_k - s_k^{(i)}\|_\infty \leq \rho, \quad \forall k \in \{0, \dots, N\} \\ & s_{k+1} = A_k s_k + B_k u_k + c_k, \quad \forall k \in \{0, \dots, N-1\} \end{aligned}$$

where the terminal constraint $s_N = s_{\text{goal}}$ is replaced by the terminal cost $(s_N - s_{\text{goal}})^\top P (s_N - s_{\text{goal}})$, and A_k, B_k, c_k come from the linearization as follow:

$$s_{k+1} = f_{\mathbf{d}}(s_k^{(i)}, u_k^{(i)}) + \frac{\partial f_{\mathbf{d}}}{\partial s}(s_k^{(i)}, u_k^{(i)})(s_k - s_k^{(i)}) + \frac{\partial f_{\mathbf{d}}}{\partial u}(s_k^{(i)}, u_k^{(i)})(u_k - u_k^{(i)}) = A_k s_k + B_k u_k + c_k$$

with

$$A_k = \frac{\partial f_{\mathbf{d}}}{\partial s}(s_k^{(i)}, u_k^{(i)}), \quad B_k = \frac{\partial f_{\mathbf{d}}}{\partial u}(s_k^{(i)}, u_k^{(i)}), \quad c_k = f_{\mathbf{d}}(s_k^{(i)}, u_k^{(i)}) - A_k s_k^{(i)} - B_k u_k^{(i)}$$

(b) Please see `cartpole_swingup_constrained.py`.

(c)

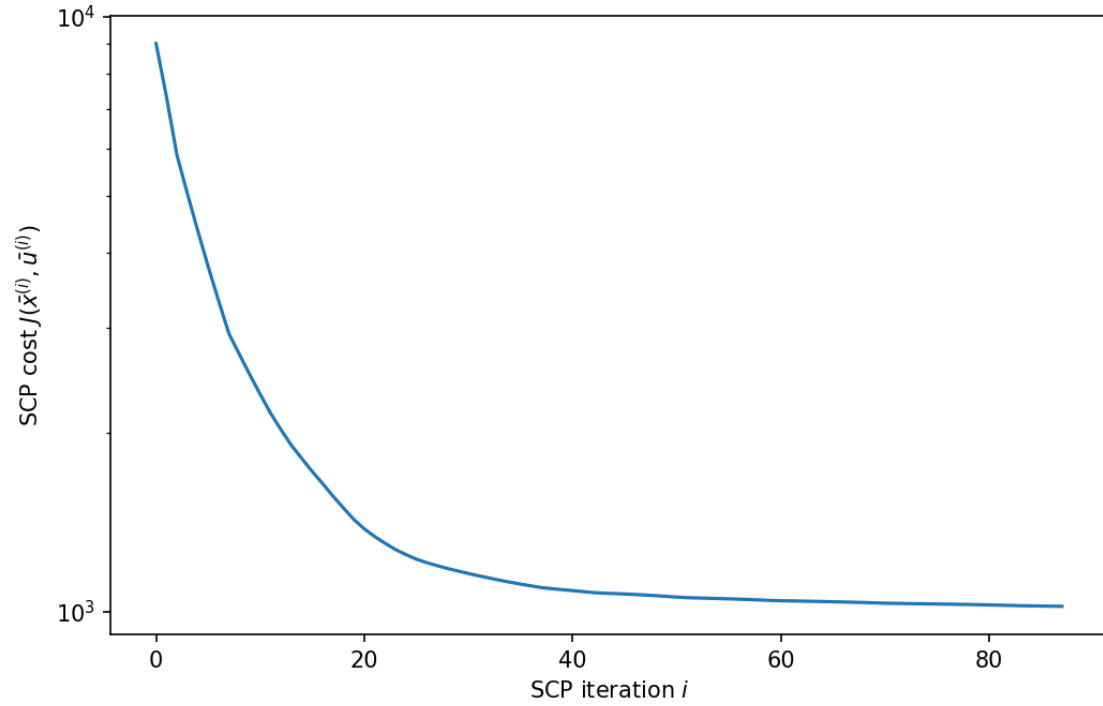
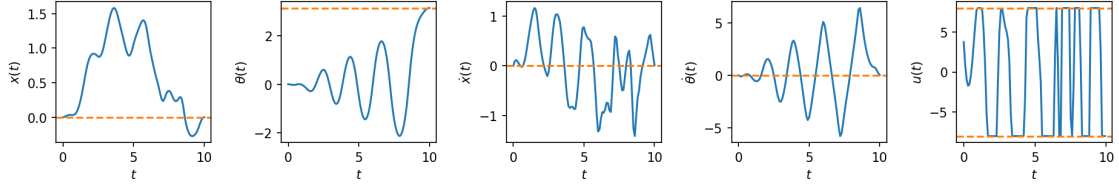
Please see `cartpole_swingup_constrained.py`.

(d)

```
[1]: %run cartpole_swingup_constrained.py
```

```
88%|          | 88/100 [01:21<00:11, 1.08it/s, objective change=0.32265]
```

SCP converged after 88 iterations.



The pendulum just reaches s_{goal} at the end of the horizon but does not stay there. A natural scheme is to use SCP to generate the swing-up trajectory, then switch to a closed-loop LQR controller linearized around the upright equilibrium to keep the pendulum there indefinitely.